Chapter 7

Solutions to exploration 7.1

1. Explanatory: Narrow/wide angle, Response: Time it takes to run to second base
2. Gather data on the time it takes players to run to second base by watching a bunch of games, then seeing if people who rounded first base tend to get to second base faster than those who took a narrow angle. We can't draw cause and effect conclusions because there are many confounding variables that haven't been controlled for through random assignment.
3. Randomly assign the 20 players to the two conditions (wide vs. narrow angle)
4. Random assignment should place some fast runners and some slow runners in each group, so runner speed is not a confounding variable
5. Have each player run both ways (once wide and once narrow)
6. Randomly decide whether each player runs wide or narrow first
7. Difference in times
8. Pair runners who have the same time in the 100 yard dash. Randomly assign one runner from each pair to run the wide angle and one to run the narrow angle.
9. In the repeated measures design the individuals in the pair are exactly the same because they are the same person, this is the best study design.
10. Responses will vary. In marriages, are males typically older than females? Pairs=married couples, pairing likely effective because couples will have experienced similar life events. Explanatory variable is (husband/wife) and response is age.
11. Responses will vary. Suppose that you want to investigate whether teenagers in the United Kingdom have read more *Harry Potter* books on average than teenagers in the United States. Explanatory variable: country (UK or US), response variable: number of books read. There is no obvious way to pair teenagers from the two countries.
12. If you are comparing different treatments on lab animals, say beetles, and the response variable can only be found by killing the beetle (i.e., measuring something that is internal) then repeated measures is not possible.
13. Use the time of the 100-yard dash from the 8 players who rounded first with the narrow angle and match those times to the closest 8 times out of the 12 players who ran the wide angle. That will give 8 matched pairs. We didn't assign the angle, but we can still try to compare the most similar runners to each other.

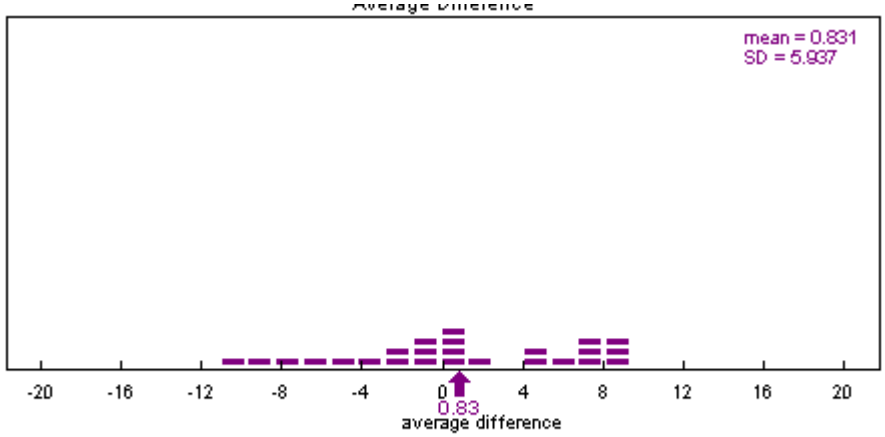
Solutions to Exploration 7.2

1. Answers will vary.
2. Because it is the same people in both groups, and so the groups are not "independent."
3. The contextual explanation is that if there are any "carry-over" effects of any kind (e.g., heart rate higher for second exercise compared to first because the heart rate was already high; people don't try as hard for second because they are tired, etc.) then the random assignment of which treatment is first will help to balance those effects between the two conditions. More broadly, random assignment of the paired treatment orders will help to more clearly suggest

that the treatment (exercise type) is the cause of the change in heart rate and not other confounding variables (being tired, initial heart rate, etc.).

4. Null: The long-run mean difference in heart rates between the two exercises is zero
   Alt: The long-run mean difference in heart rates between the two exercises is not zero
   <or>
   Null: $\mu_d=0$
   Alt: $\mu_d\neq0$

   From this point forward we use the JJvsBicycle dataset in the solutions
5. Average difference is 11.955 beats p/minute
6. What it might look like in a class of 25 students. The observed statistic (11.955) is out in the tail.



Number of Samples: 25

7. Jumping jacks mean=114.65, sd=19.57; Bicycle kicks mean=102.68, sd=20.66; mean diff=11.955, sd=20.10.
8. (after randomization—answers will vary) JJ mean=110.68, sd=21.68, Bicycle kicks mean=106.64, sd=20.15, mean diff=4.045, sd=5.828
   a. Actual mean diff is more extreme.
   b. 11.955 > 4.045
9. a. 100 dots
   b. The mean of the differences from a single trial (re-randomization of the order of the observations)
10. a. Bell shaped curve,
    b. centered at zero—null hypothesis says no difference, we are simulating the null hypothesis
    c. sd ≈ 4.9, this represents-the spread or variability of the simulated mean differences
11. 11.955 is in the tail of the distribution meaning it is unlikely to occur by chance alone
12. Example results: p-value ≈ 0.006
13. We have strong evidence that the average difference in heart rate comparing jumping jacks to bicycle kicks is different from 0. This is not a random sample so generalizing to any population is questionable—maybe students at this particular college?

14. 11.955/4.9 = 2.4. The standardized statistic is larger than 2, and so gives strong evidence against the null (as does the p-value)
15. 11.955 +/- 2×4.9 = 11.955 +/- 9.8 = (2.2, 21.8)
16. We are 95% confident that the long-run mean diff in heart rates (JJ-bicycle) is between 2.2 and 21.8 beats per minute. In other words, we are 95% confident the long-run mean heart rate after JJ is between 2.2 and 21.8 beats per minute faster than after bicycles.
17. Difference in means, average=-0.06, SD=6.34, p-value=0.062
18. The SD is LARGER when analyzing as independent samples
19. The p-value is smaller when analyzing as paired samples

Solutions to exploration 7.3

1. a. Because he paired identical cards and then randomly assigned one of each kind to the two different kinds of auctions
   b. 2 cards
   c. Type of auction (Dutch or first-price sealed bid); categorical
   d. Price paid; quantitative
2. Null hypothesis: The long-run mean price paid for magic cards in a Dutch auction is the same as in a sealed bid auction
   Alt hypothesis: The long-run mean price paid for magic cards is different in the two different auction types
3. The difference in long-run mean prices paid between the two different types of auctions; μ_d
4. Null: $\mu_d=0$
   Alt: $\mu_d \neq 0$
5. A. 5.16, b. 4.78, c. mean=0.384, SD=0.675, d. Using 1000 repetitions of the random assignment of the order of the observations, we found a p-value=0. The p-value was found by counting what proportion of the simulated mean differences were as extreme as 0.384 (in either direction, beyond)
6. Yes, there are 88 pairs without strong skewness in the distribution of differences.
7. A. t=5.34, the observed statistic is more than 5 sd's from the mean
   b. the p-value should be very small because the observed statistic is very large,
   c. p-value < 0.0001, similar to what we got from simulation
8. a. (0.240, 0.527)
   b. We are 95% confident that on average the selling price of a Magic card in a Dutch auction is more than the selling price of a card in a first-price sealed bid auction, by between $0.240 and $0.527.
   c. Yes, it does not include 0.
9. We have very strong evidence of a difference in average prices paid using the two different auction formats because our p-value was <0.0001; with the price of the Dutch auction being, on average, between $0.24 and $0.53 higher (95% confident) than the first-price sealed bid auction. Cause-and-effect is possible here because cards were randomly assigned to treatment

groups (type of auction) so we have convincing evidence that the type of auction causes the difference in average prices. We cannot generalize this result beyond magic cards--- there is no guarantee that these 88 pairs of magic cards are somehow representative of “all” magic cards.